

Question: 1

Update your project documentation by adding a 'Known Issues & Limitations' section based on feedback you received during the peer review session?

Answer: • 1. IP Address Exhaustion Risk in Dynamic Zones:

- Issue: The current /24 subnet mask limits each department/VLAN to a maximum of 254 usable IP addresses.

- Limitation: During high-traffic marketing events or peak guest arrivals, this range can quickly exhaust, requiring a future upgrade to a larger /23 subnet range.

- 2. Inter-VLAN Routing Bottleneck on the Core Switch:

- Issue: All internal data routing between different departments relies entirely on a single Layer 3 Core Switch configuration.

- Limitation: Simultaneous heavy data transfers between the Tech and Marketing teams could strain the Core Switch's CPU processing power, causing minor latency across the local network.

- 3. Budget Inflation Due to High Hardware Redundancy:

- Issue: The high-availability redundancy plan introduces duplicate premium components like dual routers and dual Cisco core switches.

- Limitation: Deploying identical secondary backup hardware significantly increases the initial Bill of Materials (BOM) cost, which might exceed the budget constraints of smaller co-working start-ups.

- 4. Physical Security Risks of Exposed Access Points:

- Issue: Wireless Access Points are mounted in open, shared co-working spaces and corridors for optimal signal distribution.

- Limitation: Peer feedback highlighted that these public setups are vulnerable to unauthorized physical tampering or malicious ethernet cable extraction by visitors.

- 5. Session Disruption for Long-Term Guest Users:

- Issue: The security policy enforces a strict 4-hour session timeout on the Guest VLAN captive portal to prevent bandwidth hoarding.

- Limitation: Verified clients or guests booking full-day passes will experience frequent disconnections, forcing them to re-authenticate multiple times throughout the day, hurting user experience.

Question: 2

Revise your presentation slides to clearly highlight at least two improvements or changes you made after receiving peer feedback.

Hint: Use before-and-after visuals or bullet points to show what was changed and why.?

Answer: Improvement 1: Enhancing Physical Security for Access Points

- Before Feedback: Wireless Access Points (APs) were simply planned to be mounted openly on corridor ceilings with exposed cabling. Peers pointed out that this left the network vulnerable to physical tampering.
- After Feedback (Revision): Updated the layout specifications to mandate secure lockable wall-mount enclosures for all Access Points.
- Reason for Change: To prevent physical tampering, ensure unauthorized users cannot unplug the network equipment, and protect the physical integrity of the devices in shared public zones.

Improvement 2: Resolving Guest Captive Portal Session Expiry Friction

- Before Feedback: The security slide stated a strict 4-hour session timeout for everyone on the Guest VLAN network, requiring users to log in repeatedly.
- After Feedback (Revision): Modified the security policy to implement a tiered session duration rule. Standard daily guests retain the 4-hour limit, but verified clients with premium day passes are assigned an 8-hour extended session lease via a different access code.

- Reason for Change: To maintain robust network security while drastically improving the user experience for long-term paying clients who require uninterrupted connectivity during the workday.

Question: 5

Write out and rehearse concise answers to three tough questions you were asked during the peer panel, focusing on how you addressed each concern in your revised design?

Answer: • Panel Question 1 (On Physical Security):

- "Since Access Points are mounted in shared open areas, how do you prevent visitors from physically tampering with the network lines or hardware?"

• Rehearsed Concise Answer: "That is a critical point. In our revised design, we replaced standard open mounts with industrial-grade lockable anti-tamper enclosures for all communal Access Points. Additionally, all ethernet drops are routed through solid metal conduits inside the wall, entirely eliminating exposed cabling layer access."

• Panel Question 2 (On Project Budget & Redundancy Costs):

- "Your plan uses duplicate Cisco Core switches and dual edge routers. Isn't this excessive and overly expensive for a small co-working start-up?"

• Rehearsed Concise Answer: "While dual hardware layout increases initial capital expenditure, network downtime is directly tied to financial losses for paying co-working tenants. We mitigated this by utilizing cost-effective multi-WAN gateway routers (TP-Link ER605) alongside standard Cisco units to achieve full link and hardware redundancy without completely ballooning the Bill of Materials."

• Panel Question 3 (On User Experience vs. Security Rules):

- "A strict 4-hour captive portal session timeout will frustrate long-stay users. How do you balance this security restriction with customer satisfaction?"

- Rehearsed Concise Answer: "To address this, we transitioned from a rigid blanket rule to a tiered session management policy. Standard hourly visitors retain the secure 4-hour lease window, whereas verified daily pass holders are issued extended access credentials that extend their uninterrupted connection lease to an 8-hour workday window.

Question: 4

Refine your system architecture diagram or flowchart to incorporate suggestions from your peers, ensuring all major components and data flows are clearly labeled?

Answer: • 1. High Availability WAN Layer (Gateway Redundancy):

- Implementation: Upgraded from a single router baseline to a Dual-WAN Load-Balancing and Failover gateway.

- Data Flow: Primary internet traffic defaults through ISP-1 (Fiber Gateway Router A). In case of a line failure, Virtual Router Redundancy Protocol (VRRP) shifts the active traffic path instantly to Backup Gateway Router B via ISP-2.

- 2. Redundant Core Switching Matrix:

- Implementation: The central backbone now deploys a dual-switch configuration consisting of Core Switch A (Active Master) and Core Switch B (Hot Standby) interconnected via a high-bandwidth Trunk Aggregation Link.

- Loop Prevention: Rapid Spanning Tree Protocol (RSTP) maps alternate block paths across both units to eliminate duplicate broadcast storms while ensuring instantaneous sub-second hardware failover routes.

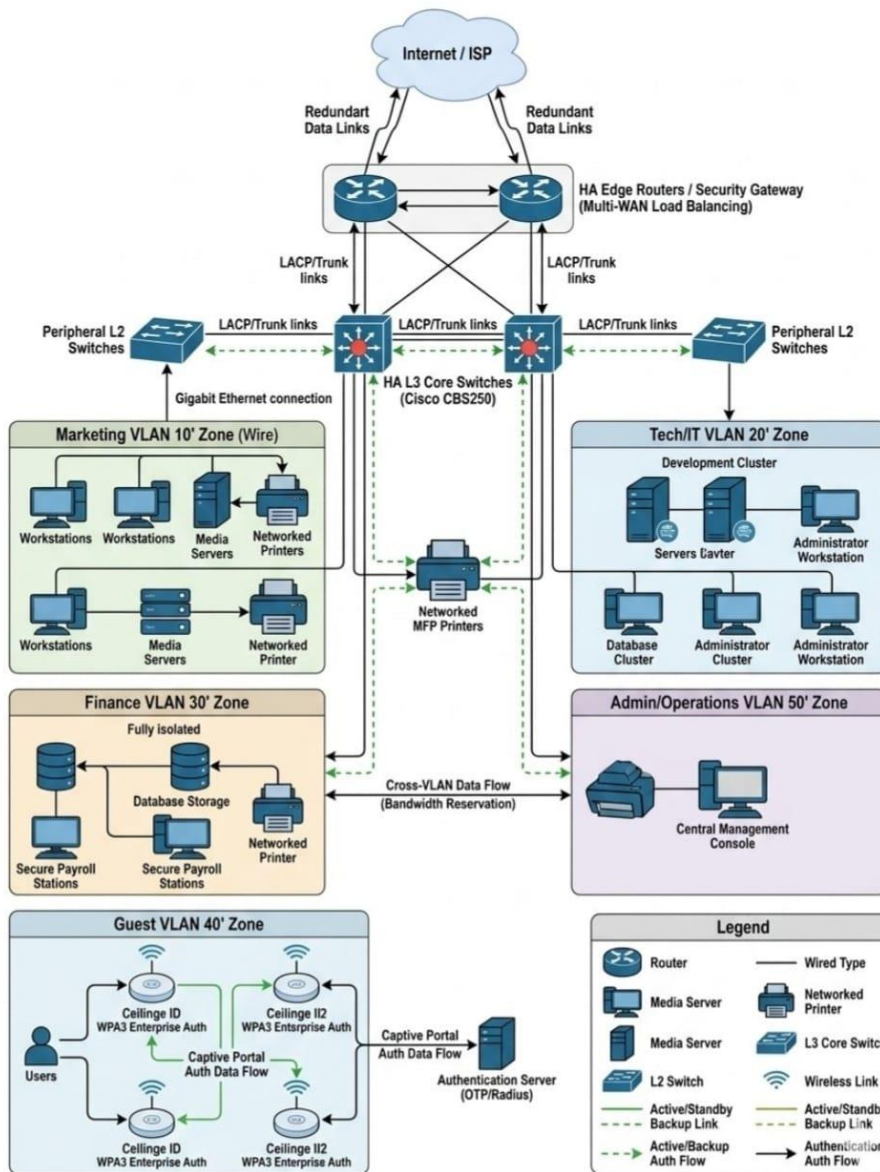
- 3. Isolated VLAN Broadcast Domains:

- Clean layout distribution across 4 explicit segments to completely separate data vectors:

- VLAN 10 (Marketing): Heavy localized upload conduits dedicated to media processing rigs.

- VLAN 20 (Tech / IT): Direct routing architecture provisioning development environments and internal sandbox servers.
- VLAN 30 (Finance): Secure, sandboxed accounting subnets blocked off from default cross-communication layers.
- VLAN 40 (Guest Network Management): Dynamic client infrastructure utilizing an outer-ring internet-only gateway.
- 4. Enhanced Edge Layer Physical Hardening:
 - Implementation: All floor-distributed Wireless Access Points (APs) are explicitly labeled with Industrial Lockable Anti-Tamper Enclosures fed exclusively by internal wall conduits to completely eliminate external physical penetration vectors.

Title: Refined Co-Working Space High-Availability Network Architecture



Question: 5

Record a 2-minute video or audio clip practicing your project pitch, aiming to confidently explain the main features and improvements made post-review?

Answer: Introduction (0:00 - 0:20)

- *Greeting & Objective:* "Good day, everyone. Today, I am pitching our refined high-availability network architecture designed for a modern co-working space."
- *Core Goal:* "Our primary objective was to build a robust, secure, and multi-tenant environment that caters seamlessly to three distinct departments: Marketing, Tech, and Finance, while supporting a dynamic guest zone."

Main System Features (0:20 - 0:55)

- *Traffic Segmentation (VLANs):* "At the heart of our design is strict logical isolation. We implemented dedicated VLANs: VLAN 10 for Marketing's heavy media needs, VLAN 20 for Tech development sandboxes, and a highly sandboxed VLAN 30 to protect sensitive Financial transactions."
- *Hardware Selection:* "To make this feasible within realistic market constraints, our Bill of Materials centers on industry-reliable components like the multi-WAN TP-Link ER605 edge router and premium Layer 3 Cisco CBS250 core switches."

Post-Review Improvements & Refinements (0:55 - 1:40)

- *Eliminating Single Points of Failure:* "Following valuable feedback from our peer review session, we significantly hardened the infrastructure. First, we addressed system vulnerability by introducing a dual-router gateway using Virtual Router Redundancy Protocol (VRRP) and stacked dual-core switches with Rapid Spanning Tree Protocol (RSTP) to guarantee sub-second automated failovers."
- *Physical Layer Hardening:* Second, we tackled physical security concerns. All communal Wireless Access Points are now housed inside industrial-grade lockable anti-tamper enclosures with internal wall conduits to completely eliminate local tampering risks."
- *Optimized User Experience:* "Lastly, we refined our network security policy. We replaced a rigid 4-hour guest portal cutoff with a tiered lease structure, extending pass-holder connections to a friction-free 8-hour workday window while preserving security logs via a Radius OTP server."